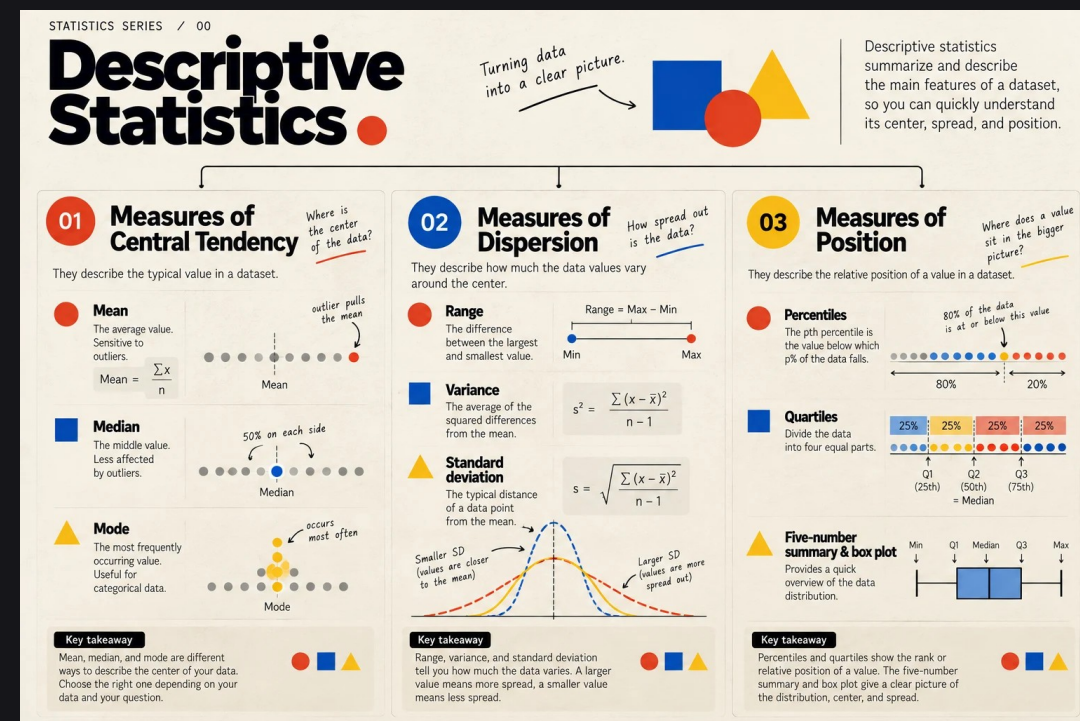




Descriptive Statistics

How to read the story hidden in data



Why descriptive statistics matter

Before any model, forecast, or dashboard, someone has to understand what the data actually looks like. Descriptive statistics are that first step: they summarize a dataset in a few numbers, so we can see its shape before we draw conclusions.

1

They make large datasets readable

A few measures can stand in for thousands of rows.

2

They stop us from trusting a single number

A mean of €64.18 hides that most customers spend between €40 and €60, and two stores can share a mean of €50 while behaving very differently.

3

They reveal unusual observations

A value like €180 stands out once we look at quartiles and box plots, and it raises the right question: is it an error or a real customer?

4

They prepare everything that follows

Choosing the right chart, model, or business conclusion depends on first knowing the center, spread, and position of the data.

Three questions every dataset can answer



CENTER

Where is the data?

Mean • Median • Mode



SPREAD

How much does it vary?

Range • Variance • Standard deviation



POSITION

Where does an observation sit?

Percentiles • Quartiles • IQR

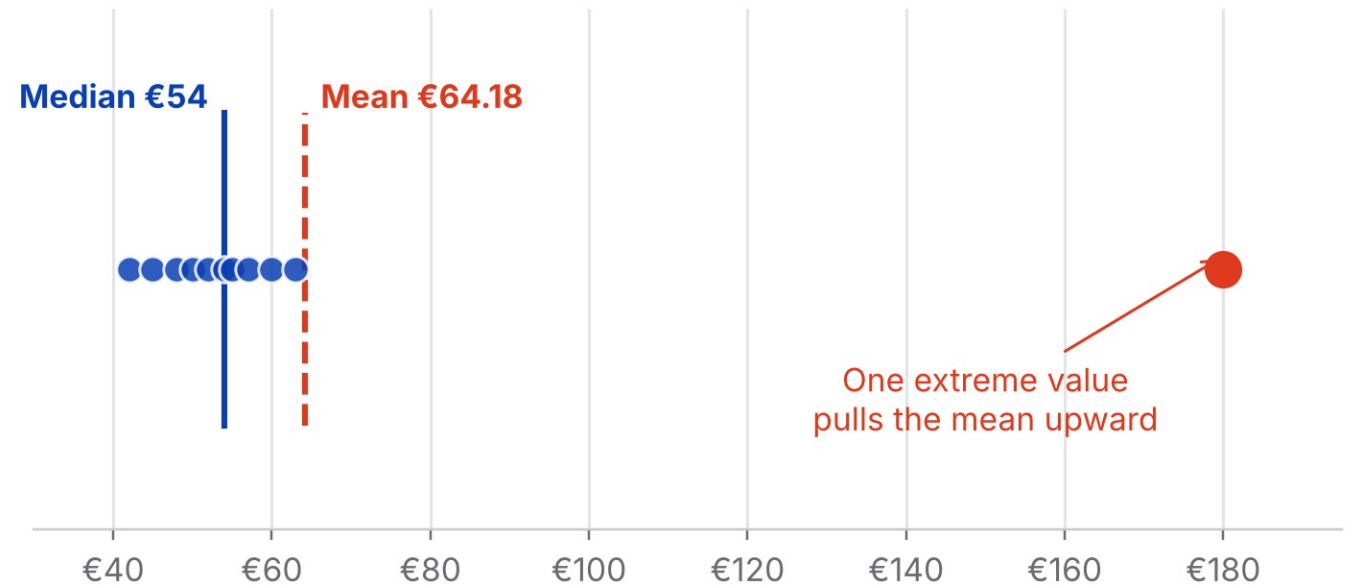
Mean — the mathematical average

Add all observations and divide by the number of observations.

$$\bar{x} = \sum x_i / n$$

Example: monthly customer spending = €42, €45, €48, €50, €52, €54, €55, €57, €60, €63, €180

Mean = **€64.18**



ANALYTICAL LESSON

The mean uses every observation, so it is sensitive to extreme values and skewed distributions.

Median — the middle of the data

Order the observations and find the middle value. With an even number of observations, average the two middle values.

For the same spending data, the median is €54.

M E D I A N

€54

M E A N

€64.18

WHY IT MATTERS

Most customers in this example spend around €40–€60, while one customer spends €180. The median remains close to the typical customer.

USE IT WHEN

The distribution is skewed or contains meaningful outliers—for example, house prices, salaries, or customer spending.

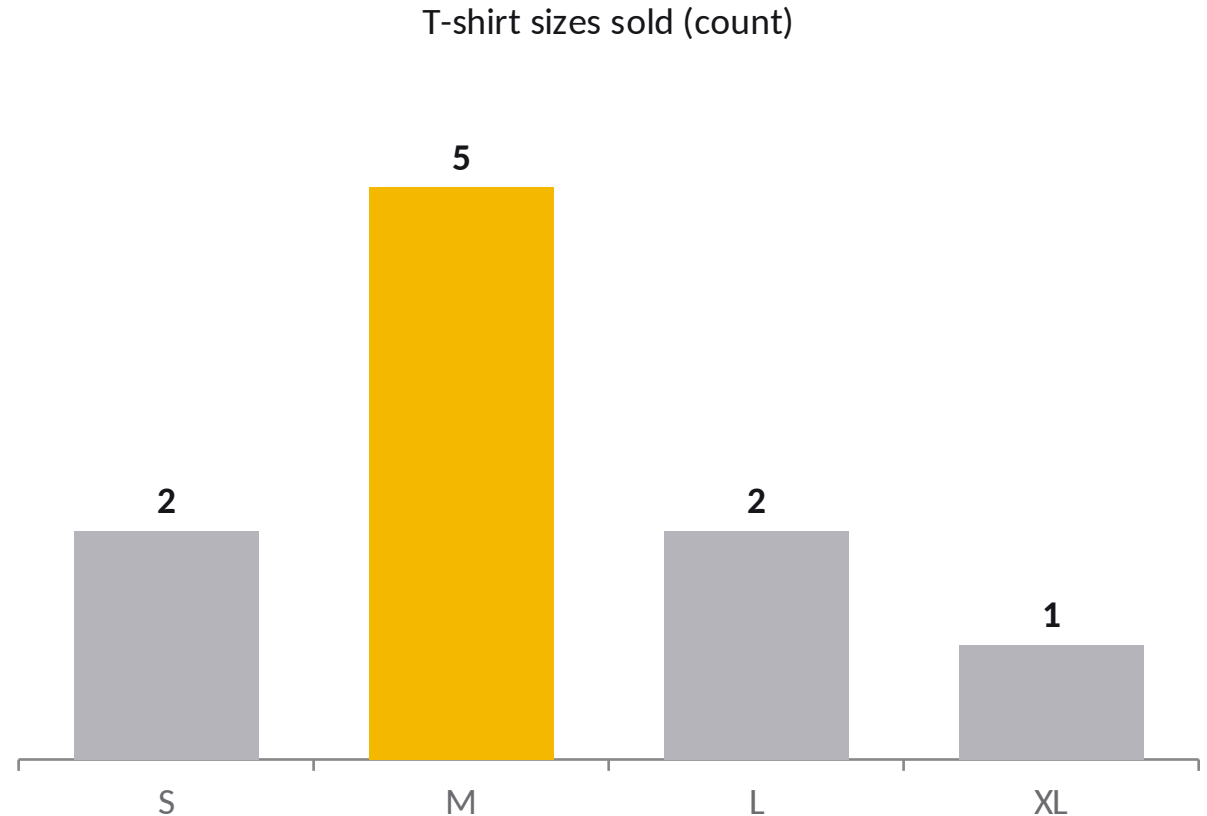
Mode — what happens most often

The mode is the most frequently occurring value. A dataset can have no mode, one mode, or multiple modes.

Example: T-shirt sizes sold = S, M, M, L, M, XL, L, M, S, M.
The mode is M.

ESPECIALLY USEFUL FOR

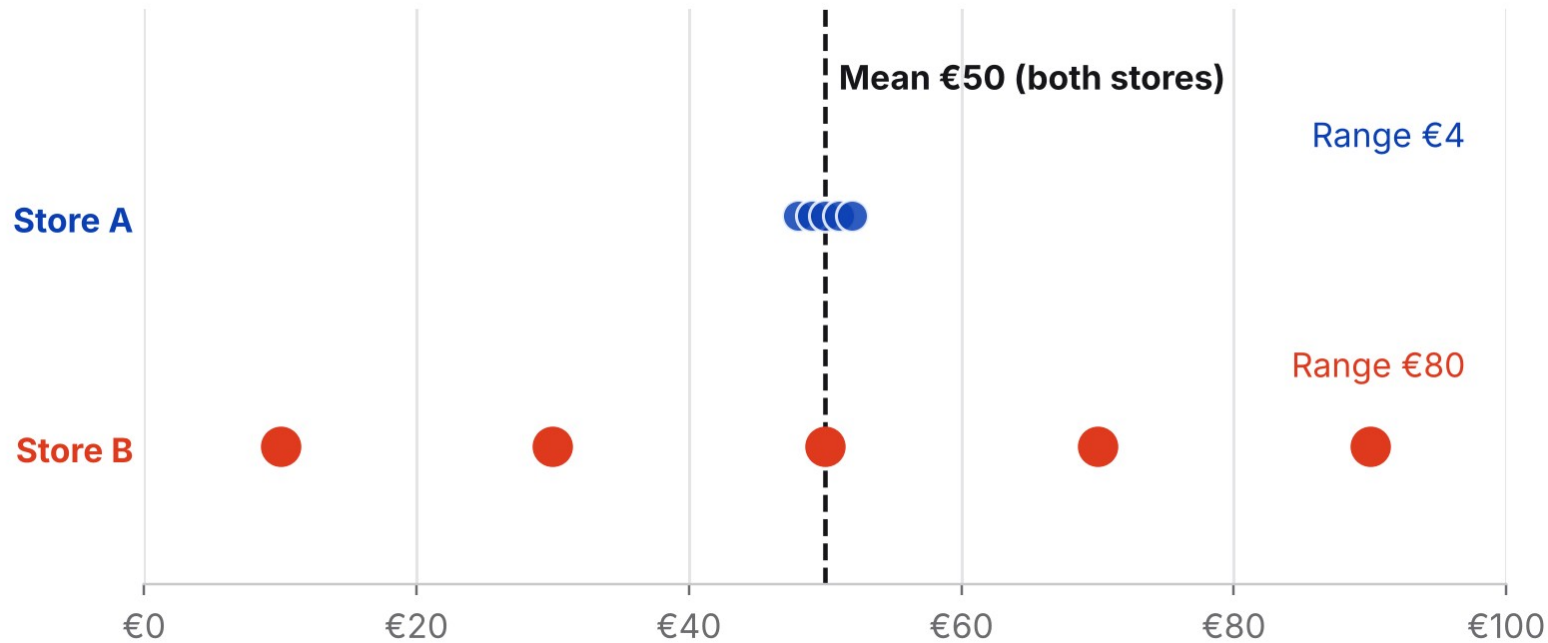
Categorical data, where an average may not make sense. For example, the most common payment method can be Credit Card.



Mean, median or mode?

Situation	Useful measure
Numeric, relatively symmetric data	Mean
Numeric data with outliers	Median
Skewed distributions	Median
Categorical data	Mode
Most common category	Mode

Same mean, very different spread



Knowing the center is not enough. Two datasets can have the same mean and completely different levels of variability.

Store A: €48, €49, €50, €51, €52

Store B: €10, €30, €50, €70, €90

Both have a mean of €50—but Store B is much more spread out.

KEY IDEA

Central tendency tells us where the data is. Dispersion tells us how much the observations vary.

Three ways to measure spread

Range

$$\text{Range} = \text{max} - \text{min}$$

Store A: €52 - €48 = €4

Store B: €90 - €10 = €80

Limitation: it depends only on the minimum and maximum, so one extreme value can change it dramatically.

Variance

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$

The average squared distance from the mean. Squaring prevents positive and negative deviations from cancelling each other out.

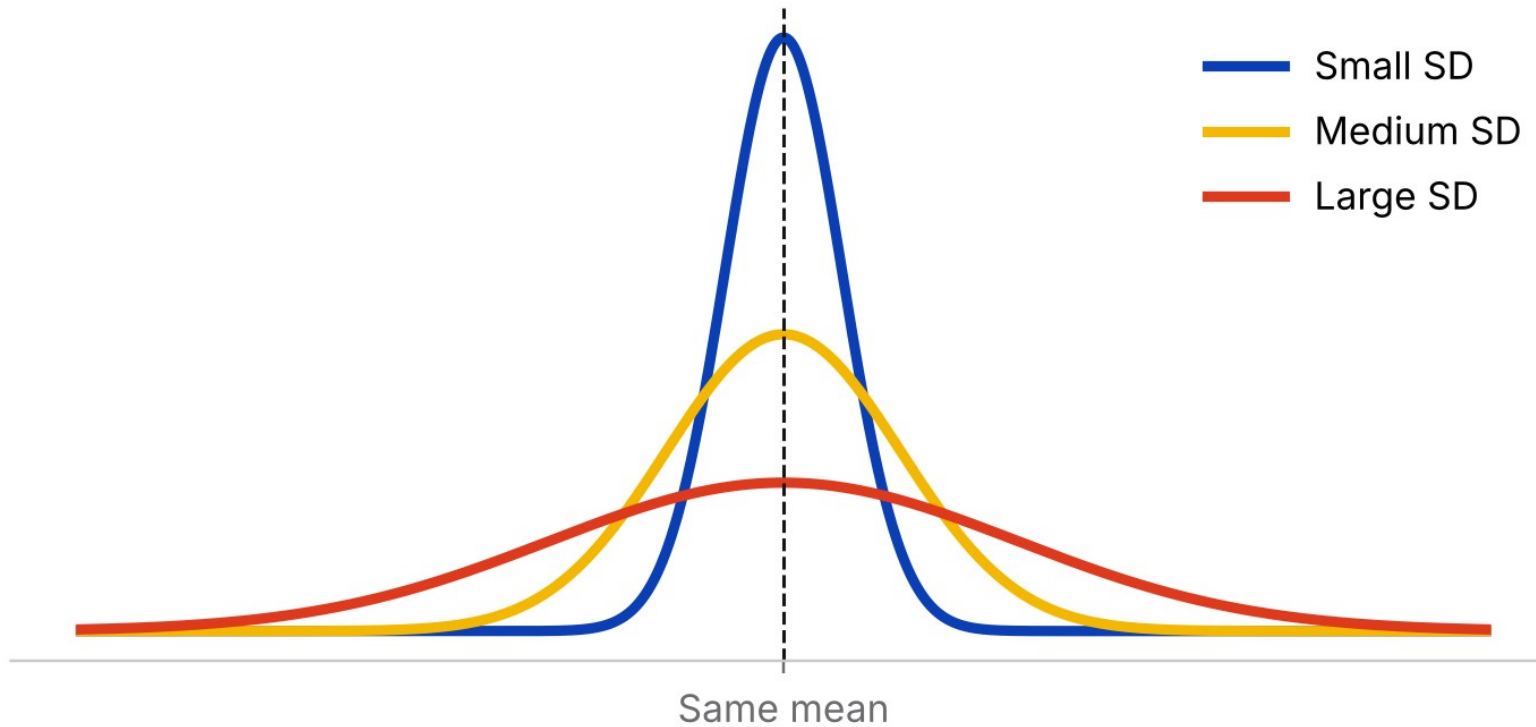
Standard deviation

$$s = \sqrt{\text{variance}}$$

The square root of variance, back in the original unit.

Dataset 2, 3, 10: mean = 5, squared deviations = 9, 4, 25, variance = 19, SD ≈ 4.36

Same mean, wider curve, greater spread



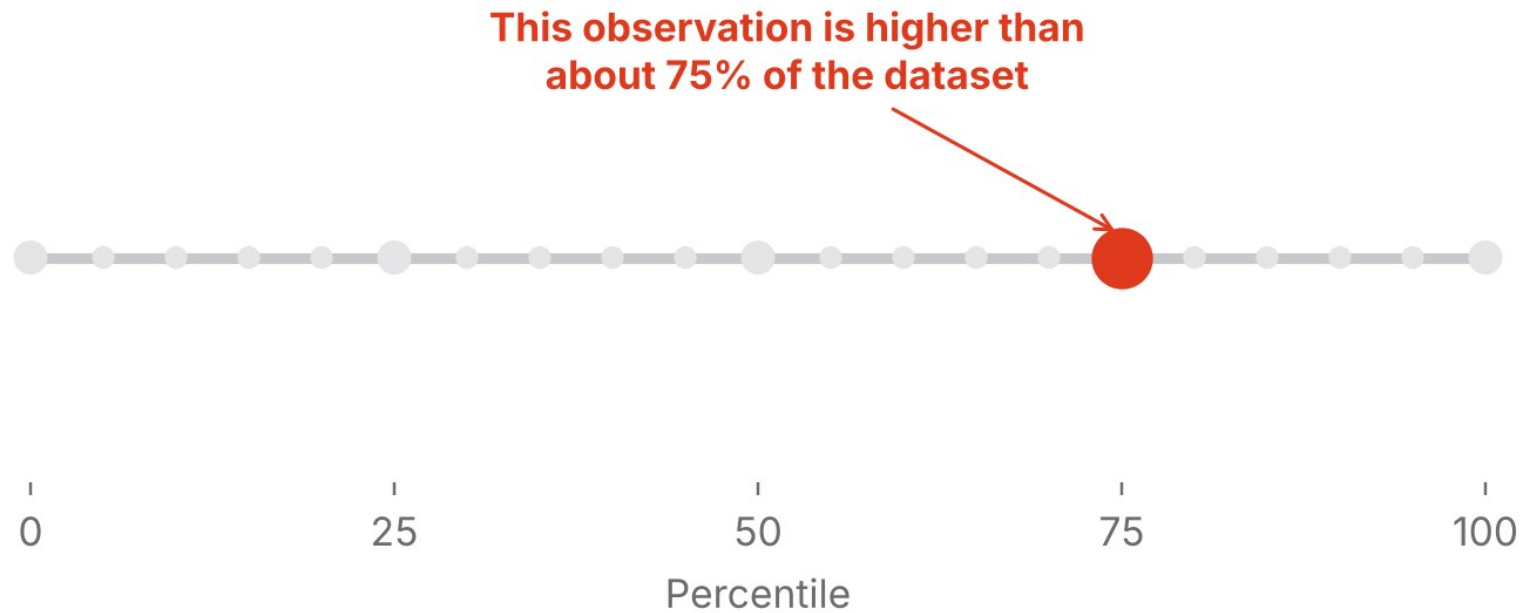
Standard deviation returns spread to the original unit of measurement. The wider the curve, the greater the spread.

BUSINESS EXAMPLE: RENTAL PRICES

Two neighborhoods may have similar average rents but very different variability. A larger standard deviation means a wider spread of rents, which matters for price predictability.

Percentiles — relative position

Percentiles divide ordered data into 100 parts. A value at the 75th percentile is higher than approximately 75% of observations in the reference dataset.

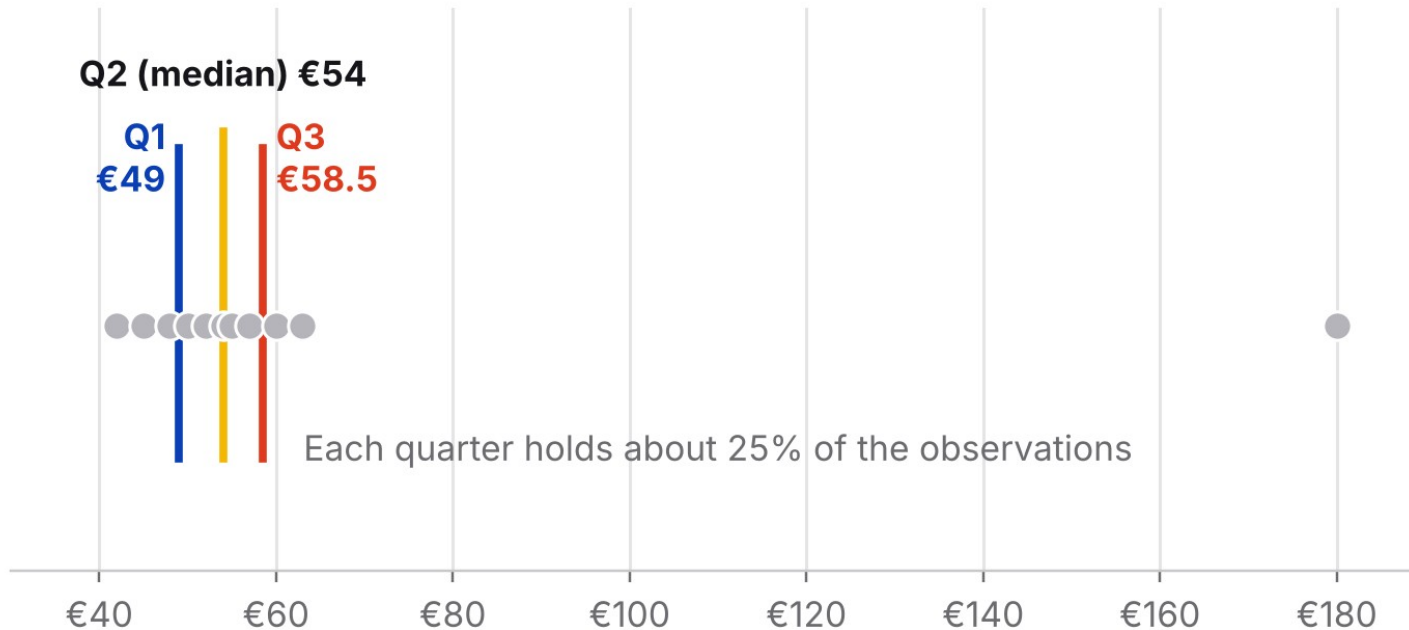


IMPORTANT DISTINCTION

The 75th percentile does not mean 75% correct. It describes relative position, not the percentage of points achieved.

Quartiles and the IQR

Quartiles divide the ordered dataset into four parts: Q1 = 25th percentile | Q2 = median | Q3 = 75th percentile.



$$\text{IQR} = Q3 - Q1$$

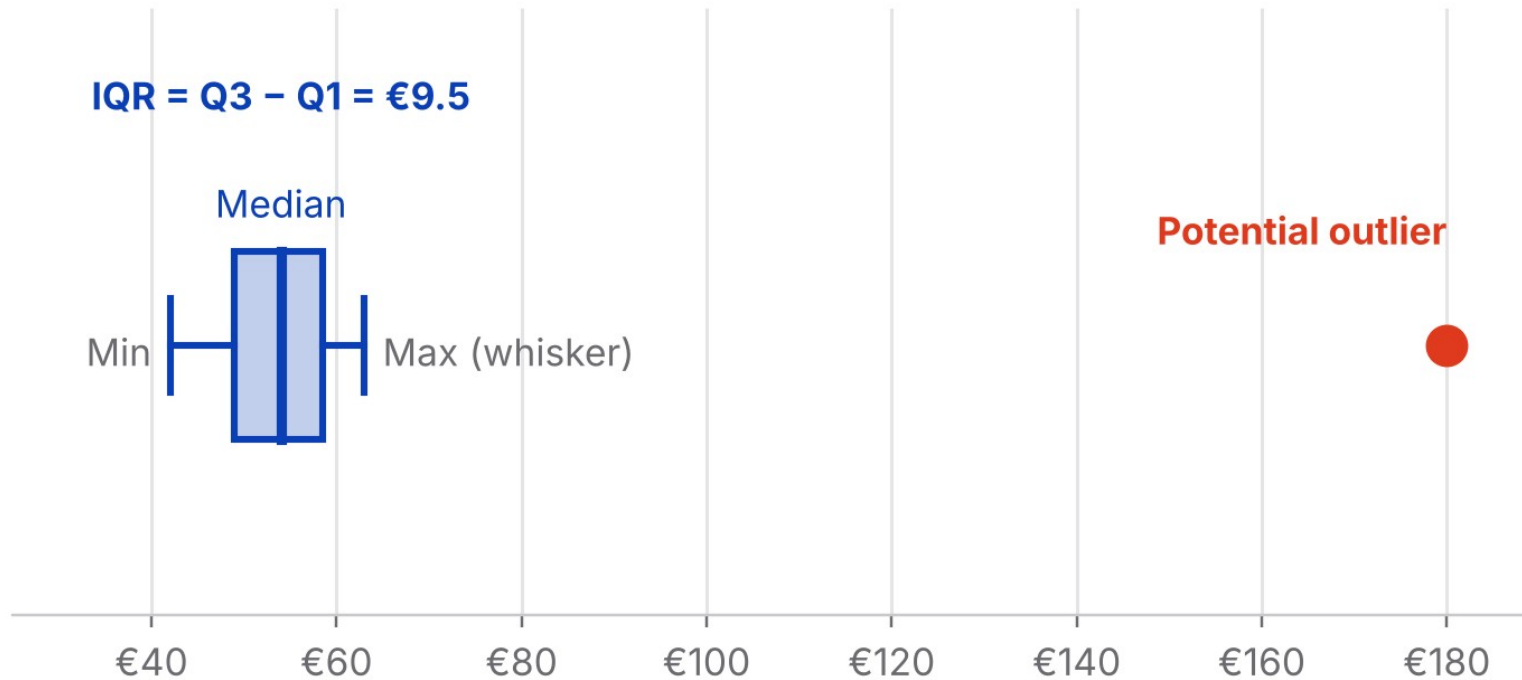
If Q1 = €35 and Q3 = €78, then IQR = €43. The middle 50% of observations occupy a €43-wide interval.

WHY IT MATTERS

The IQR is less sensitive to extreme values than the full range, making it useful for skewed data and outlier analysis.

Five-number summary and box plot

Minimum → Q1 → Median → Q3 → Maximum



The box

represents the middle 50%.

The line inside

is the median.

The whiskers

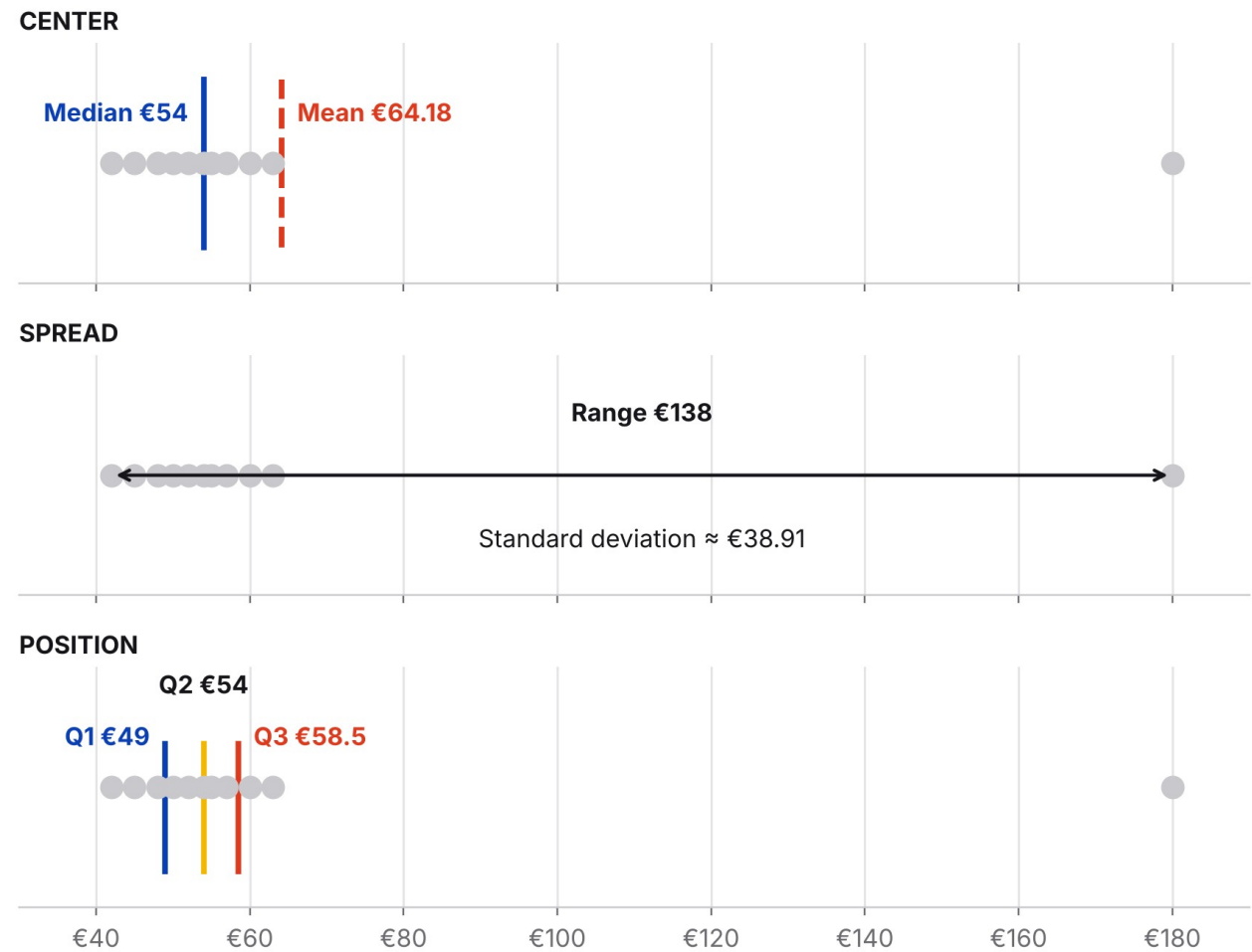
show the remaining distribution according to the chosen box-plot convention.

Putting everything together

Customer spending: 42, 45, 48, 50, 52, 54, 55, 57, 60, 63, 180

Question	Measure	What it tells us
Where is the center?	Mean / Median	Typical level and sensitivity to extreme values
What occurs most often?	Mode	Most common value or category
How much does it vary?	Range / Variance / SD	Overall and typical spread
Where does an observation sit?	Percentile / Quartile	Relative position
What is the middle 50%?	IQR	Robust measure of spread
Are there unusual observations?	IQR / Box plot	Potential outliers and distribution shape

One dataset, three lenses



CENTER

The mean and median tell us where the data is centered.



SPREAD

Range and standard deviation tell us how much it varies.



POSITION

Percentiles and quartiles tell us where observations sit within the distribution.

Final takeaway

Descriptive statistics are not just numbers to calculate. They are different lenses for looking at the same data. The mean and median tell us where the data is centered. Range and standard deviation tell us how much it varies. Percentiles and quartiles tell us where observations sit within the distribution. Together, they turn a collection of numbers into something we can interpret.

